

ANTIOESTROGENIC ACTION OF THE ALDOSTERONE ANTAGONIST CANRENOATE K IN THE RAT (ADENOHYPOPHYSIS, CERULOPLASMIN)

V. SCHREIBER, T. PŘIBYL

Laboratory for Endocrinology and Metabolism, Third Medical Clinic, Faculty of Medicine, Charles University, Prague

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Summary

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In a dose of 7.5 mg/rat/day in food, the aldosterone antagonist canrenoate K inhibited the adenohypophyseal reaction (growth, raised thyroxine-binding capacity of the adenohypophyseal proteins *in vitro*) and the ceruloplasmin reaction (elevation of the serum ceruloplasmin level) to three weeks' intramuscular administration of long-acting oestradiol benzoate in doses of 1 mg twice a week. The effect was similar to that of the antioestrogen clomiphene in a dose of 1.25 mg/rat/day in food. In combined administration of clomiphene and canrenoate K, no summation of their effect was observed. Neither canrenoate nor clomiphene affected the post-oestradiol drop in body weight, but they both potentiated the oestradiol-induced decrease in testicular weight and canrenoate potentiated the effect of oestradiol on uterine weight. It was therefore concluded that the effect of canrenoate is not of a catatonic nature, i.e. that it is not determined by increased metabolic degradation of oestradiol.

The administration of large doses of oestrogens is followed by adenohypophyseal hyperplasia (Selye et al. 1935, Burrows 1936) which may develop after a time into an adenohypophyseal tumour (Cramer and Horning 1936). The thyroxine-binding capacity of the adenohypophyseal proteins rises at the same time (Schreiber et al. 1970). Elevation of the blood ceruloplasmin level is a further proteosynthetic reaction to oestrogens (Meani and Cartel 1970, Briggs and Briggs 1971, Alias 1971, Sunderman et al. 1971). The reaction of the adenohypophysis and the increase in the blood level of ceruloplasmin (which is formed in the liver) are apparently only coincidental and not causally related, although the polyphenol oxidase activity of ceruloplasmin, and hence its ability to inactivate, *inter alia*, dopamine, does not allow complete preclusion of a causal association. By inactivating dopamine, ceruloplasmin could act at the level of hypothalamic regulation of adenohypophyseal growth. After administering oestrogen, we found a positive correlation between adenohypophyseal weight and the serum ceruloplasmin level in rats (Schreiber and Přibyl 1976).

In preceding reports we showed that various hormones and neurotropic drugs have a parallel effect on all three reactions to oestrogens, i.e. adenohypophyseal growth, raised thyroxine-binding capacity of the adenohypophyseal proteins and elevation of the ceruloplasmin level. Excess of the thyroid

hormones and testosterone inhibit them [Schreiber and Přibyl 1977]. Perphenazine inconstantly potentiates the effect of oestradiol on the adeno-hypophysis [Schreiber et al. 1972], but does not potentiate its effect on the ceruloplasmin level and an ergoline derivative inhibited both reactions of the adeno-hypophysis [Schreiber and Přibyl 1977a]. The dopaminergic blocker Pimozide, however, potentiated the effect of oestradiol both in the adeno-hypophysis and on the serum ceruloplasmin level [Schreiber and Přibyl 1976a].

Since the adeno-hypophyseal reaction to oestrogens can be inhibited by administering antioestrogens [Schreiber et al. 1972a, b], we wondered whether they would also inhibit the increase in the ceruloplasmin level. At the same time we decided to test the effect of canrenoate K, a derivative of spironolactone (an aldosterone antagonist), as Laube and Quirk (1975) described five cases of breast cancer associated with spironolactone therapy. Although these authors themselves drew attention to the unlikelihood of there being a causal association between spironolactone and breast cancer and to the fact that spironolactone suppresses the carcinogenic effect of dimethylbenzanthracene on the mammary gland in rats [Kovács and Somogyi 1970], there is no doubt that gynaecomastia frequently develops in men treated with spironolactone and its derivatives [Mann 1963, Clark 1965]. Since women with breast cancer often have a raised ceruloplasmin level [Schapira 1972, Schreiber et al. 1975], we thought it would be interesting to determine the effect of the spironolactone derivative canrenoate K on the serum ceruloplasmin level and on its reaction to oestradiol.

Spironolactone and its derivative canrenoate K are active antagonists of aldosterone (see Ross 1965) and are used in the treatment of primary and secondary aldosteronism and certain forms of hypertension. Apart from gynaecomastia (mentioned above), one of their endocrine side effects is amenorrhoea [Levitt 1970]. Gynaecomastia is evidently caused by the antiandrogenic action of spironolactone, which inhibits dihydrotestosterone binding to the cytosol and nuclear receptor [Corvol et al. 1975]. Spironolactone also reduces endogenous testosterone production, however [Menard et al. 1974] and is an active inductor of hepatic microsomal drug-metabolising enzyme systems [Feller and Gerald 1971], which is evidently the cause of its marked catatoxic effect — it inhibits the effect of various sedatives and anaesthetics [Selye et al. 1969a] and of digitoxin [Selye et al. 1969].

We therefore decided to compare the effect of the antioestrogen clomiphene and the spironolactone derivative canrenoate K in rats given large doses of oestradiol. In the given model (pituitary weight, T₄ binding to the adeno-hypophyseal proteins in vitro, the ceruloplasmin level), canrenoate K was found to have a distinct antioestrogenic effect.

Material and Methods

In two experiments, 80 male (exp. No. 1) and female (exp. No. 2) Wistar rats (Vědeňský ústav, Prague) were divided into eight groups: 1) controls 2) animals receiving oestradiol benzoate as an aqueous microcrystal suspension (Agofolin depot SPOFA), 1 mg i.m. twice a week; 3) clomiphene citrate (Gravosan SPOFA), 1.25 mg/rat/day in food; 4) canrenoate K (Aldactone Boehringer), 7.25 mg/rat/day in food; 5) oestradiol + clomiphene; 6) oestradiol + canrenoate K; 7) clomiphene + canrenoate K; 8) oestradiol + clomiphene + canrenoate K. The animals were kept in cages in groups of five (temperature $22 \pm 2^\circ\text{C}$, indirect daylight) and were fed on a standard laboratory diet [Larsen diet, Vědeňský ústav, Prague].

After three weeks' treatment the animals were exsanguinated and the adeno-hypophyses, adrenals, gonads and seminal vesicles or uteri were dissected out and weighed on a torsion balance. ^{125}I -thyroxine binding to the adeno-hypophyseal proteins *in vitro* was determined by the method described in previous reports [Schreiber and Přibyl 1976b, c] and the results were expressed as the percentage of the thyroxine dose bound by the proteins in 1 ml adeno-hypophyseal homogenate, as the number of ng thyroxine bound to the proteins in 1 mg [wet weight] adeno-hypophysis and as the number of ng thyroxine bound to 1 mg adeno-hypophyseal protein. The serum ceruloplasmin level was determined as polyphenol oxidase activity [Přibyl 1976], using a human ceruloplasmin preparation [Serva, Heidelberg] as the standard. The means and the 95% confidence limits were calculated for all the above values and the significance of the differences of the means was tested by Duncan's test [Duncan 1955].

Results

In the males (Tab. 1, Fig. 1), oestradiol markedly raised adeno-hypophyseal weight, slightly raised adrenal weight and sharply reduced the weight of the

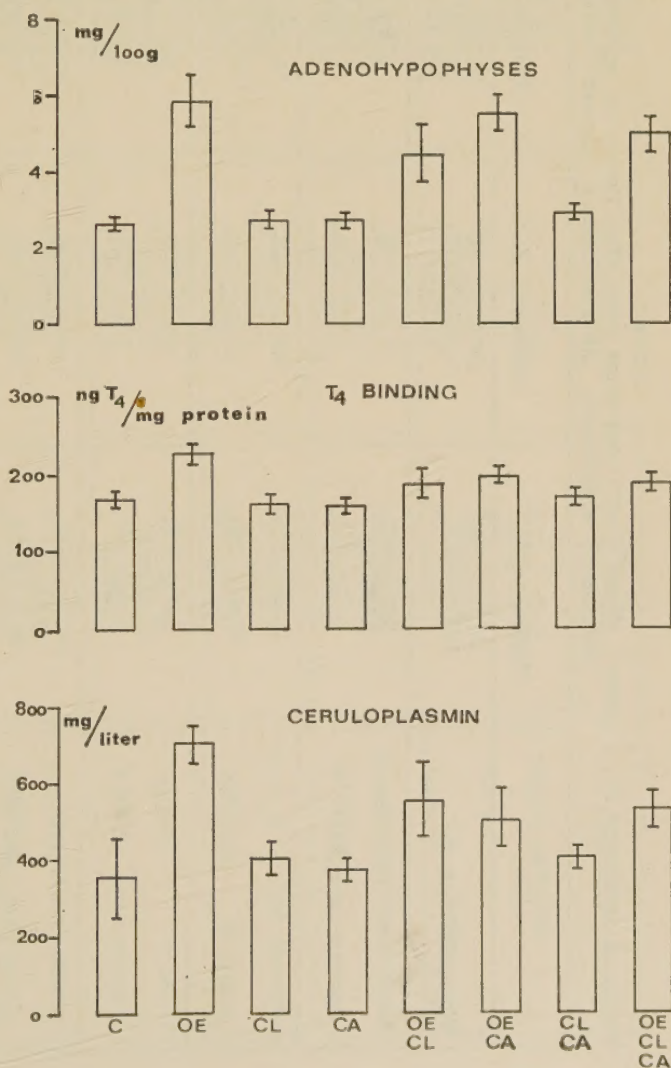


Fig. 1. Mean values + 95% confidence limits for relative adeno-hypophyseal weight, thyroxine binding to adeno-hypophyseal proteins *in vitro* (ng T₄/mg prot.) and ceruloplasmin concentration in control male rats and after administration of oestradiol (OE), clomiphen (CL) and carenatoate (CA), either or variously combined.

Table 1

Results of experiment No. 1 (males). Means $\pm$ 95% confidence limits. The figures in brackets are the numbers of groups with statistically significantly different means (Duncan's test).

Group (N)	1. Controls (10)	2. Oestradiol (10)	3. Clomiphene (10)	4. Canrenoate (9)
Body weight, g, init. final	138.50 $\pm$ 2.41 199.30 $\pm$ 12.78 (2-8)	138.00 $\pm$ 2.50 147.20 $\pm$ 10.76 (1,3,4,7)	136.50 $\pm$ 2.41 167.70 $\pm$ 8.80 (1,2,4-6,8)	137.22 $\pm$ 2.78 181.50 $\pm$ 12.03 (1-3,5-8)
Adenohypophyses mg mg/100 g	5.18 $\pm$ 0.51 (1,2,3,5,6,8) 2.59 $\pm$ 0.11 (2,5-8)	8.61 $\pm$ 1.35 (1,3-5,7) 5.84 $\pm$ 0.79 (1,3-5,7,8)	4.59 $\pm$ 0.54 (1,2,5,6,8) 2.73 $\pm$ 0.26 (2,5,6,8)	4.94 $\pm$ 0.44 (2,5,6,8) 2.73 $\pm$ 0.23 (2,5,6,8)
Adrenals, mg mg/100 g	48.50 $\pm$ 4.64 (5,6,8) 23.00 $\pm$ 1.98 (2-8)	42.90 $\pm$ 6.09 (3,5-8) 29.07 $\pm$ 3.04 (1,5-8)	52.70 $\pm$ 7.00 (2,5,8) 31.42 $\pm$ 3.65 (1,4,5,8)	47.78 $\pm$ 7.03 (5,7,8) 26.23 $\pm$ 3.00 (1,3,5-8)
Testes, g g/100 g	2.47 $\pm$ 0.13 (2,3,5-8) 1.25 $\pm$ 0.08 (2,5,6,8)	0.88 $\pm$ 0.22 (1,3-8) 0.59 $\pm$ 0.14 (1,3-6,8)	1.81 $\pm$ 0.40 (1,2,4-6,8) 1.07 $\pm$ 0.22 (2,5,6,8)	2.46 $\pm$ 0.50 (2,3,5-8) 1.38 $\pm$ 0.29 (2,5,6,8)
Seminal vesicles, mg mg/100 g	206.70 $\pm$ 32.58 (2,3,5-8) 104.08 $\pm$ 15.88 (2,3,5-8)	69.50 $\pm$ 12.88 (1,3,4) 46.86 $\pm$ 6.27 (1,4)	112.90 $\pm$ 38.59 (1,2,4-6,8) 66.58 $\pm$ 21.91 (1,4,8)	238.67 $\pm$ 55.94 (2,3,5-8) 130.46 $\pm$ 28.15 (2,3,5-8)
¹²⁵ I-T4 binding % ng T4/mg w.w. ng T4/mg protein	20.41 $\pm$ 0.77 (2,5,6,8) 12.75 $\pm$ 0.48 (2,5,6,8) 171.50 $\pm$ 8.26 (2,5,6,8)	27.00 $\pm$ 1.98 (1,3-8) 16.88 $\pm$ 1.24 (1,3-8) 232.40 $\pm$ 16.33 (1,3-8)	20.23 $\pm$ 1.58 (2,5,6,8) 12.64 $\pm$ 0.99 (2,5,6,8) 171.10 $\pm$ 12.63 (2,5,6,8)	19.35 $\pm$ 1.20 (2,5,6,8) 12.10 $\pm$ 0.75 (2,5,6,8) 162.56 $\pm$ 11.83 (2,5,6,8)
Ceruloplasmin mg/l	357.65 $\pm$ 91.50 (2,5,6,8)	701.10 $\pm$ 44.40 (1,3-8)	401.00 $\pm$ 32.95 (2,5,6,8)	376.56 $\pm$ 38.29 (2,3,6,8)

Table 1 (Continuation)

Group	5. Oestradiol + Clomiphene [10]	6. Oestradiol + Canrenoate [10]	7. Clomiphene + Canrenoate [10]	8. Oestradiol + Clomiphene + Canrenoate [10]
Body weight, g, init. final	138.50 ± 2.41 154.30 ± 11.28 (1.3, 4.7)	139.00 ± 2.26 149.30 ± 7.35 (1.3, 4.7)	138.00 ± 2.50 166.30 ± 8.24 (1.2, 4-6.8)	137.00 ± 1.85 150.50 ± 7.37 (1.3, 4.7)
Adenohypophyses, mg mg/100 g	6.87 ± 1.29 (1-4.6, 7) 4.41 ± 0.74 (1-4.6, 7)	8.21 ± 0.93 (1.3-5.7) 5.49 ± 0.46 (1.3-5.7)	4.78 ± 0.50 (2.5, 6.8) 2.87 ± 0.24 (1.2, 5.6, 8)	7.49 ± 1.04 (1.3, 4.7) 4.96 ± 0.56 (1-4.7)
Adrenals, mg mg/100 g	62.10 ± 6.76 (1-4.6, 7) 40.38 ± 4.27 (1-4.6, 7)	51.90 ± 5.49 (1.2, 5.8) 34.81 ± 3.42 (1.2, 4.5, 8)	55.10 ± 4.90 (1.2, 4.5, 8) 33.12 ± 2.49 (1.2, 4.5, 8)	66.10 ± 7.25 (1-4.6, 7) 43.85 ± 3.72 (1-4.6, 7)
Testes, g g/100 g	0.64 ± 0.12 (1-4.7) 0.41 ± 0.07 (1-4.7)	0.58 ± 0.11 (1-4.7) 0.39 ± 0.07 (1-4.7)	1.60 ± 0.67 (1.2, 4-6.8) 0.93 ± 0.37 (5.6, 8)	0.56 ± 0.08 (1-4.7) 0.37 ± 0.05 (1-4.7)
Seminal vesicles, mg mg/100 g	68.90 ± 9.00 (1.3, 4) 44.85 ± 5.62 (1.4)	72.40 ± 11.61 (1.3, 4) 48.41 ± 6.96 (1.4)	77.60 ± 31.99 (1.4) 45.79 ± 17.51 (1.4)	66.50 ± 13.30 (1.3, 4) 44.29 ± 8.71 (1.3, 4)
¹²⁵ I-T4 binding % ng T4/mg w.w. ng T4/mg protein	23.35 ± 1.29 (1-4.7, 8) 14.59 ± 0.86 (1-4.7, 8) 192.10 ± 12.56 (1-4.7)	23.84 ± 0.90 (1-4.7, 8) 14.90 ± 0.56 (1-4.7, 8) 199.70 ± 7.34 (1-4.7)	20.20 ± 1.35 (2.5, 6.8) 12.62 ± 0.84 (2.5, 6.8) 170.80 ± 11.66 (2.5, 6.8)	21.96 ± 1.01 (1-7) 13.73 ± 0.63 (1-7) 190.10 ± 8.52 (1-4.7)
Ceruloplasmin mg/l	557.10 ± 91.54 (1-4.7)	497.70 ± 69.68 (1-4.7)	407.70 ± 27.47 (2.5, 6.8)	531.00 ± 56.48 (1-4.7)

testes and seminal vesicles. Thyroxine binding to the adenohipophyseal proteins *in vitro* and the serum ceruloplasmin level both rose abruptly after oestradiol (the latter to double the control value). Clomiphen raised adrenal weight, lowered the weight of the seminal vesicles and did not affect either thyroxine binding to the adenohipophyseal proteins *in vitro* or the ceruloplasmin level. Canrenoate K mildly raised adrenal weight, raised the weight of the seminal vesicles and did not affect the other parameters. The combination of clomiphen and canrenoate produced no further changes, except that it inhibited the post-canrenoate increase in the weight of the seminal vesicles. Clomiphen depressed the effect of oestradiol on adenohipophyseal weight, but potentiated its effect on adrenal weight; the effect on the seminal vesicles

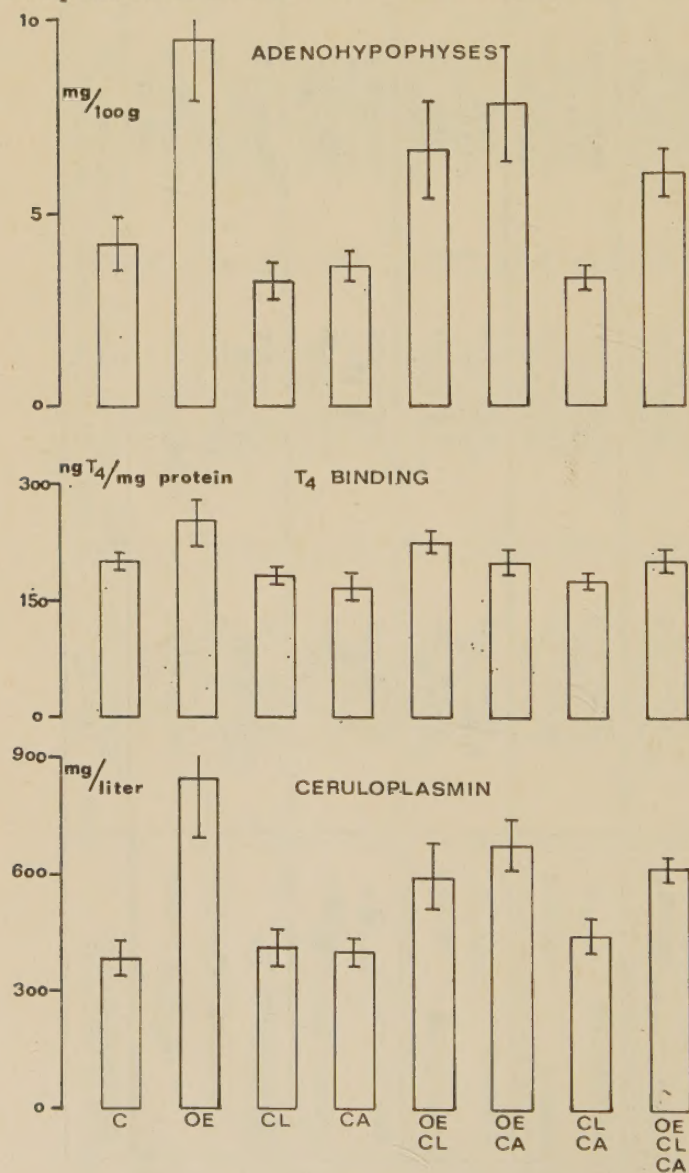


Fig. 2. Mean values + 95% confidence limits for relative adenohipophyseal weight, thyroxine binding to adenohipophyseal proteins *in vitro* (ng T₄/mg prot.) and ceruloplasmin concentration in control female rats and after administration of oestradiol [OE], clomiphen [CL] and canrenoate [CA], either singly or variously combined.

Table 2

Results of experiment No. 2 (females). Means $\pm$ 95% confidence limits. The figures in brackets are the numbers of groups with statistically significantly different means (Duncan's test).

Group (N)	1. Controls (10)	2. Oestradiol (10)	3. Clomiphen (10)	4. Canrenoate (10)
Body weight, g, init. final	152.50 $\pm$ 3.04 185.00 $\pm$ 12.59 (2,3,5-8)	155.50 $\pm$ 2.64 159.60 $\pm$ 10.11 (1,4,6,8)	152.50 $\pm$ 4.21 154.40 $\pm$ 9.03 (1,4)	155.50 $\pm$ 4.60 183.00 $\pm$ 16.01 (2,3,5-8)
Adenohypophyses, mg	7.83 $\pm$ 1.57 (2,3,5-7) 4.23 $\pm$ 0.73 (2,3,5-8)	15.07 $\pm$ 2.50 (1,3-8) 9.52 $\pm$ 1.63 (1,3-8)	4.94 $\pm$ 0.62 (1,2,4-6,8) 3.22 $\pm$ 0.47 (1,2,5,6,8)	6.59 $\pm$ 0.97 (2,3,5-8) 3.60 $\pm$ 0.41 (2,5,6,8)
Adrenals, mg	53.80 $\pm$ 5.31 (2,3,7,8) 29.15 $\pm$ 2.70 (5)	47.80 $\pm$ 3.64 (1,3,5,7) 30.2 $\pm$ 3.55 (5)	42.20 $\pm$ 4.44 (1,2,4,5) 27.56 $\pm$ 3.61 (5)	52.90 $\pm$ 8.95 (3,7) 29.13 $\pm$ 4.69 (5)
Ovaries, mg	93.70 $\pm$ 15.41 (3,7) 50.49 $\pm$ 6.93 (5-7)	91.40 $\pm$ 16.56 (3,7) 58.08 $\pm$ 12.29 (3-5,7)	68.00 $\pm$ 6.82 (1,2,4-6,8) 44.12 $\pm$ 3.97 (2,5-8)	81.10 $\pm$ 9.96 (3,5,7) 46.47 $\pm$ 5.88 (2,5-8)
Uteri, mg	304.50 $\pm$ 76.17 (2,3,5-8) 166.73 $\pm$ 45.18 (2,5,6-8)	524.70 $\pm$ 91.34 (1,3,4,7) 333.33 $\pm$ 70.26 (1,3,4,6,7)	187.40 $\pm$ 62.38 (1,2,5,6,8) 122.75 $\pm$ 45.34 (2,5,6,8)	247.10 $\pm$ 67.66 (2,5-8) 134.31 $\pm$ 30.4 (2,5,6,8)
¹²⁵ I-T4 binding % ng T4/mg w.w. ng T4/mg protein	23.18 $\pm$ 0.86 (2-5,7) 14.48 $\pm$ 0.53 (2,4,5,7) 197.88 $\pm$ 9.26 (2,4,5,7)	29.17 $\pm$ 3.64 (1,3,4,6-8) 18.23 $\pm$ 2.28 (1,3,4,6-8) 251.70 $\pm$ 31.96 (1,3,4,6-8)	21.63 $\pm$ 1.54 (1,2,4,5,6,8) 13.52 $\pm$ 0.96 (2,4-6,8) 183.80 $\pm$ 12.02 (1,2,4,5,7)	19.35 $\pm$ 1.60 (1-3,5,6,8) 11.89 $\pm$ 0.95 (1-3,5,6,8) 164.10 $\pm$ 14.98 (1-3,5,6,8)
Ceruloplasmin, mg/l	379.80 $\pm$ 39.97 (2,5-8)	837.90 $\pm$ 157.54 (1,3-8)	408.60 $\pm$ 43.96 (2,5,6,8)	400.50 $\pm$ 38.38 (2,5,6,8)

Table 2 (Continuation)

Group (N)	5. Oestradiol + Clomiphene (10)	6. Oestradiol + Canrenoate (10)	7. Clomiphene + Canrenoate (10)	8. Oestradiol + Clomiphene + Canrenoate (10)
Body weight, g, init. final	158.00 ± 4.82 150.80 ± 11.52 (1,4)	155.00 ± 5.05 148.80 ± 4.95 (1,2,4,7)	157.00 ± 4.82 158.80 ± 3.70 (1,4,6)	155.50 ± 4.28 152.40 ± 5.04 (1,2,4)
Adenohypophyses, mg mg/100 g	9.96 ± 1.99 (1-4,7,8) 8.62 ± 1.24 (1-4,7)	11.60 ± 2.28 (1-4,7,8) 7.84 ± 1.52 (1-4,7,8)	5.29 ± 0.48 (1,2,4-6,8) 3.33 ± 0.26 (1,2,5,6,8)	9.04 ± 1.06 (2-4,6,7) 5.93 ± 0.65 (1-4,6,7,)
Adrenals, mg mg/100 g	52.20 ± 3.58 (2,3,7,8) 35.00 ± 3.88 (1-4,7,8)	48.00 ± 8.27 (3,7) 32.24 ± 5.45	43.50 ± 4.35 (1,2,4,5) 27.24 ± 2.50 (5)	46.50 ± 5.41 (1,5,) 50.53 ± 3.49 (5)
Ovaries, mg mg/100 g	102.70 ± 11.90 (3,4,5-8) 68.35 ± 7.23 (1,3,4,6-8)	89.80 ± 8.82 (3,5,7) 60.32 ± 5.39 (1,3-5,7)	60.00 ± 5.28 (1-6,8) 37.83 ± 3.49 (1-6,8)	86.40 ± 9.86 (3,5,7) 56.65 ± 5.87 (3-5,7)
Uteri, mg mg/100 g	477.00 ± 52.92 (1,3,4,6,7) 319.72 ± 43.69 (1,3,4,6,7)	604.80 ± 89.08 (1,3,4,5,7) 407.22 ± 61.27 (1-5,7,8)	174.50 ± 23.97 (1,2,4-6,8) 109.62 ± 13.94 (1,2,5,6,8)	505.50 ± 99.54 (1,3,4,7) 332.63 ± 67.24 (1,3,4,6,7)
¹²⁵ I-T4 binding, % ng T4/mg w.w. ng T4/mg protein	26.53 ± 1.30 (1,3,4,6-8) 16.58 ± 0.81 (1,3,4,6-8) 225.10 ± 13.56 (1,3,4,6-8)	23.45 ± 1.55 (2-5,7) 14.66 ± 0.97 (2-4,7) 199.40 ± 14.82 (2-4,5,7)	20.27 ± 0.91 (1,2,5,6,8) 12.67 ± 0.57 (1,2,5,6,8) 174.40 ± 7.12 (1,2,5,6,8)	23.45 ± 1.77 (2-5,7) 14.64 ± 1.11 (2-5,7) 200.00 ± 16.63 (2,4,5,7)
Ceruloplasmin mg/l	589.50 ± 84.04 (1-4,6,7)	674.10 ± 60.52 (1-5,7,8)	437.40 ± 43.96 (1,2,5,6,8)	606.60 ± 34.73 (1-4,6,7)

was unaltered and the effect on the testes was potentiated. Clomiphen also markedly inhibited the increase in thyroxine binding to the adenohipophyseal proteins and elevation of the serum ceruloplasmin level after the oestrogen. Canrenoate (like clomiphen) potentiated the effect of the oestrogen on testicular weight, did not modify (as distinct from females) its effect on adenohipophyseal weight, but markedly inhibited the post-oestrogen increase in thyroxine binding to the adenohipophyseal proteins and in the ceruloplasmin level. The combination of clomiphen and canrenoate produced no further changes in their effect on the action of oestradiol.

In females (Tab. 2, Fig. 2), oestradiol greatly raised adenohipophyseal and uterine weight, thyroxine binding to the adenohipophyseal proteins *in vitro* and the ceruloplasmin level. Clomiphen slightly lowered adenohipophyseal weight (this effect in females was already known) and uterine weight and mildly reduced the thyroxine-binding capacity of the adenohipophyseal proteins. Apart from a mild decrease in thyroxine binding to the adenohipophyseal proteins, no changes were found after canrenoate. Clomiphen and canrenoate combined reduced the weight of the adenohipophysis, ovaries and uterus, slightly lowered thyroxine binding to the adenohipophyseal proteins *in vitro*, but mildly raised the ceruloplasmin level. The effect of oestradiol was inhibited by both clomiphen (adenohipophyseal weight distinctly, uterine weight, thyroxine binding to the adenohipophyseal proteins and the ceruloplasmin level slightly) and canrenoate (adenohipophyseal weight, thyroxine binding to the adenohipophyseal proteins, ceruloplasmin). Canrenoate somewhat potentiated the effect of the oestrogen on uterine weight, however. The combined administration of clomiphen and canrenoate produced no further changes.

Discussion

Comparison of the effects of clomiphen and canrenoate clearly showed that canrenoate imitated all the effects of clomiphen and that it thus behaves in the given experimental system like an antioestrogen and inhibits the effects of oestradiol. The only exception was testicular weight, where canrenoate (though again like clomiphen) potentiated the effect of oestradiol. Canrenoate likewise potentiated the positive effect of oestradiol on uterine weight. The changes are summed up in Tab. 3, in which the similarity of the effect of clomiphen and canrenoate is clearly manifest.

The antioestrogenic action of canrenoate is thus another hormonal side-effect of spironolactone type substances supplementing their previously known antialdosterone and antiandrogenic effects (see the introduction). The mechanism of the antioestrogenic action of canrenoate is unknown, but competition for oestrogen receptors in the adenohipophysis or the hypothalamus seems to be the most feasible explanation. It is evidently not a case of peripheral interaction (catatoxic effect, increased metabolic degradation of oestrogens), since canrenoate did not modify the effect of oestradiol on body weight or the weight of the seminal vesicles and actually potentiated its negative effect on testicular weight. The antioestrogenic action of canrenoate is also pronounced as regards the effect of oestradiol on the ceruloplasmin level; this is evidence against the possibility, mentioned in the introduction, of a relationship between spironolactone therapy and the development of breast cancer, although it does not entirely preclude it.

Table 3

Comparison of modifications of the effect of oestradiol by clomiphen and canrenoate K
 — = inhibition, + = potentiation, O = no change.

Effect of oestradiol	Clomiphen	Canrenoate K
Body weight decrement	O	O
Increase in adenohip. weight	—	— [females only]
Raised T4 binding to adenohip. proteins	—	—
Elevation of ceruloplasmin level	—	—
Decrease in weight of seminal vesicles	O	O
Decrease in testicular weight	+	+
Increase in uterine weight	O	+

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Prof. MUDr. V. SCHREIBER, DrSc., Laboratory for Endocrinology and Metabolism, U nemocnice 1, 128 21 Prague 2.